

FTTx

Fiber-to-the-x is a family of technology that uses fiber optic cables to bring really high speed internet to the 'x'. The 'x' may refer to B-Building, H-Home, C-Curb, D-Desk, P-Premises, N-Node. The 'x' refers to the end point of the connection which utilises the fiber optic cable. The devices that come after the 'x' get connection via the gateway which is established at 'x'. Generally, there is a problem with signal attenuation, which means the longer distance the cable has to cover the more interference creeps into the signal. The attenuation is combated using either repeaters at regular intervals which regenerate the signal or by using cables technology that suffer less from attenuation or by a combination of both. Fiber optic cables are not as prone to attenuation like phone lines or other co-axial cables, hence, it offers a much greater bandwidth over a much greater distance. In India, BSNL and MTNL have already started rolling out this new technology.

Power-line Internet

Using an approach similar to Dial-up, Power-Line Internet makes use of an existing infrastructure to bring the internet to people's homes. However, this isn't as popular as the other technologies known to us as the transmission speeds are pretty low, usually ranging between 256 kbits/s to 2.7 Mbits/s. One major drawback of this technology was that the internet signal could not pass through certain elements of the power system and required extra hardware to be installed depending upon the number of connections. This meant that in a more populated region the number of extra hardware installation would rise significantly and would not be worth the extra expenditure given the paltry speeds obtained via this network.

WiFi

WiFi is a lot more common technology these days with implementations in all sorts of hardware like mobile phones, televisions, refrigerators, printers, laptops, tablets etc. The IEEE designation for WiFi is 802.11. Over the years there have been numerous improvements to the protocol and newer revisions have been brought out with the latest being 802.11ac. WiFi is somewhat limited in range with interference and line of sight issues. There are different signal bands on which transmission occurs, namely, 2.4 GHz and 5 GHz. The 2.4 GHz band can go through walls and thus goes a much greater distance but a lot of devices use the 2.4 GHz band and thus interference from those devices knocks down your transmission speed, 5GHz on the other hand is